NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

TSCT(DT), Midterm Examination Fall 2025

Time: 90 minutes **Platform & Architecture for Data Science (CT-370) – A** **Max Marks: 20**

Note:

- ***Solve in chronological order.***
- ***Understanding the question is the part of exam.***

Q1) [CLO-1] **[10 Marks]**

a) ***Explain*** Data Science Life Cycle in detail. (5)

b) ***Differentiate*** between traditional data and Big Data. (5)

Q2) [CLO-2] **[5 Marks]**

Analyze sub divide by inspecting how it is carried out. Also examine its need through two real-world scenarios. (5)

Q3) [CLO-3] **[5 Marks]**

Construct the end-to-end data flow in Uber's data pipeline along with its roles, and show how these stages ensure real-time decision-making. (5)

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TSCT(DT), Midterm Examination Fall 2025

'ime: 90 minutes **Platform & Architecture for Data Science (CT-370) – B** **Max Marks: 20**

lote:

- ***Solve in chronological order.***
- ***Understanding the question is the part of exam.***

)1) [CLO-1] **[10 Marks]**

a) ***Differentiate*** Platforms vs. Tools vs. Environments with examples. (5)

b) ***Explain*** the 5 vs of big data. (5)

)2) [CLO-2] **[5 Marks]**

Examine the differences between OLTP (Online Transaction Processing) and OLAP (Online Analytical Processing) systems, also analyze how encryption is applied in both to ensure data security, and provide two real-world scenarios. (5)

)3) [CLO-3] **[5 Marks]**

Illustrate the core components and real-world instances of MODERN DATA STACK? (5)

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TSCT(DT), Midterm Examination Fall 2025

Time: 90 minutes **Fundamentals of Data Science (CT-368) – A** **Max Marks: 20**

Note: Smart watch and phones are strictly prohibited.

Q1a) What is Data Science? **Explain** the role of a Data Engineer, Data Analyst, Data Architect, and Data Scientist. Also list the important skills and tools required for each role. **CLO-1 5 marks**

Q1b) **Explain** what is the difference between Normalization and Standardization in Data Science? Explain each method with formula. **CLO-1 5 marks**

The heights (in inches) of four students are given below:

Student	Height (in inches)
1	64
2	70
3	72
4	68

(a) Normalize the dataset using Min-Max Normalization to the range [0,1].

(b) Standardize the dataset using Z-score Standardization.

Q2a) **Differentiate** between Regression and Classification in Machine Learning. **CLO-2 5 marks**

Q2b) Below is a confusion matrix from a binary classifier: **CLO-2 5 marks**

	Predicted Positive	Predicted Negative
Actual Positive	30 TP	10 FN
Actual Negative	5 FP	55 TN

Calculate Accuracy, Precision, Recall, and F1-Score. **Interpret** which metric would be more important if this model is used for disease detection.

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TSCT(DT), Midterm Examination Fall 2025

e: 90 minutes **Fundamentals of Data Science (CT-368) – B** **Max Marks: 20**

Note: Smart watch and phones are strictly prohibited.

Q1a) **Explain** is Data Science Process and its Key steps. Differentiate between datafication and digitalization. **CLO-1 5 marks**

Q1a) **Explain** the role of a Data Engineer, Data Analyst, Data Architect, Machine Learning Engineer, and Data Scientist. Also list the important skills and tools required for each role. **CLO-1 5 marks**

Q2a) **Differentiate** between Linear Regression and Logistic Regression. Analyze how Logistic Regression can be applied for predicting student performance. **CLO-2 5 marks**

Study Hours(X)	Exam Result(Y)
2	0
3	0
4	0
5	1
6	1
7	1
8	1

Q2b) **Analyze** which attribute (Weather or Humidity) would be the better root node for constructing a Decision Tree using the Information Gain criterion. Given the dataset for playing football: **CLO-2 5 marks**

Weather	Humidity	Play Football
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Rainy	Normal	Yes
Rainy	High	No

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me: 90 minutes | **Introduction to Tools and Techniques for Data Science (CT-265) – A** | **Max Marks: 20**

Note: Attempt all questions. Answer all parts of a question in a row. State your assumptions clearly.

Q1) Discuss the significance of data in personalized recommendations (e.g., Netflix, Spotify). How does data improve user experience? Give 2 examples. **[CLO-1] [5 Marks]**

Q2) Analyze the given dataset:[CLO-2] [10 Marks]

ID	Income	Purchases
1	20	2
2	35	3
3	50	5
4	80	12
5	120	30

Tasks:

a) Compute Min–Max scaling for **Income** to [0,1]. (2)
b) Standardize **Purchases** using Z-score (population). (2)
c) Transform **Purchases** with log(1+Y). (2)
d) Bin **Income** into 3 equal-width bins, label Low/Medium/High. (2)
e) Calculate the correlation coefficient (Pearson's r) between **Income** and **Purchases**. Interpret the relationship. (2)

Q3) You are provided the dataset of monthly sales and revenue.**Analyze** the trend. Does revenue follow sales Identify one mismatch.**[CLO-2] [5 Marks]**

Month	Sales	Revenue
Jan	100	1200
Feb	80	950
Mar	150	1800
Apr	200	2400
May	170	1900

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TSCT/TSCT(AI)/TSCT(DT)/TSCT(CR)/TSCT(GA), Midterm Examination Fall 2025

90 minutes

Software Engineering (CT-365) – A

e: Attempt all questions.

Max Marks: 20

Q1)

Construct the use case diagram and sequence diagram of the following system.

1) **ATM system** [Marks: 5] [CLO-3]

Q2)

a): **Compose** the requirement scenario of **Online Exam System.** [Marks: 5] [CLO-2]

b): **Propose** the Functional and Non-Functional Requirements of the **Online Exam System.**

Q3)

a): **Relate and Contrast** Agile and Dev-ops culture. [Marks: 5] [CLO-1]

b): **Interpret** the agile models for the following scenario:

i. Showing all the work, stages, visible boards, and swim-lanes makes problems obvious — bottlenecks, overloaded stages, delays.

ii. When customers or product owners are willing to be very involved / available. On-site customer or frequent feedback is a big requirement.

Q4)

Suppose a bank wants a new **Online Banking System** (web + mobile), with features like account management, money transfer, loan application, etc. The requirements are only partially known; there are high security **risks**, regulatory constraints, scalability concerns etc. [Marks: 5] [CLO-2]

Build the above scenario using the most appropriate process model and **plan** the activities.

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90 minutes

Software Engineering (CT-365) – B

Attempt all questions.

Max Marks: 20

Q1)

Construct the use case diagram and sequence diagram of the following system. **[Marks: 5] [CLO-3]**

1) **Online shopping system**

Q2)

a): **Compose** the requirement scenario of **E-Ticket Booking.** **[Marks: 5] [CLO-2]**

b): **Propose** the Functional and Non-Functional Requirements of the **E-Ticket Booking System.**

Q3)

a): **Demonstrate** the Requirement Engineering Process. **[Marks: 5] [CLO-1]**

b): **Interpret** the agile models for the following scenario:

i. A project involves short development cycles of 2–4 weeks, with working software delivered at the end of each cycle.

ii. Developers focus on simplicity in design and continuous integration. The team practices pair programming, test-driven development, and collective code ownership.

Q4)

A hospital wants to implement a new **Patient Management System** with well-defined requirements (patient registration, appointment scheduling, medical records, billing). Since requirements are clearly understood and stable; however this system continuously require **validation and testing. Plan** the activities and **construct** this system with most appropriate process model. **[Marks: 5] [CLO-2]**

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TSCT(AI)/TSCT(DT) Midterm Examination Fall 2025

Time: 90 minutes **Artificial Intelligence & Expert Systems (CT-361) – A** **Max Marks: 20**
***Note:* Attempt All Questions**

Question 1 [CLO-1] **Marks: 05**
Define Intelligence, Utility-Based Agent, State Space, Successor Function, and Inference.

Question 2 [CLO-1] **Marks: 05**
What is an Environment in Artificial Intelligence and list its different types? State which types are most suitable for a Self-Driving Car and an Email Spam Filter, with reasons.

Question 3 [CLO-2] **Marks: 05**
Interpret the initial state, goal test, successor function, and cost function for each of the following problems. Formulate your answer precisely enough to be implemented.
a. A country is divided into regions, and each region must be colored using one of four colors {Red, Blue, Green, Yellow}. No two neighboring regions can have the same color.
b. A 3-foot-tall monkey is in a room where bananas are suspended from the 8-foot ceiling. The room contains two stackable, movable, and climbable 3-foot-high crates.

Question 4 [CLO-2] **Marks: 05**
Illustrate the graph based on the given edges and explain how Breadth-First Search (BFS), Depth-First Search (DFS), and Uniform Cost Search (UCS) would find a path from Arad to Bucharest.
Graph Edges (with costs): Arad–Zerind(75), Arad–Sibiu(140), Zerind–Oradea(71), Sibiu–Fagaras(99), Sibiu–Rimnicu Vilcea(80), Fagaras–Bucharest(211), Rimnicu Vilcea–Pitesti(97).

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Time: 90 minutes **Artificial Intelligence & Expert Systems (CT-361) – B** **Max Marks: 20**
***Note:* Attempt All Questions**

Question 1 [CLO-1] **Marks: 05**
Define Artificial Intelligence, Model-Based Agent, Branching Factor, State Space, and Rationality.

Question 2 [CLO-1] **Marks: 05**
What is an Environment in Artificial Intelligence and list its different types? State which types are most suitable for a Traffic Light Simulation and a Weather Forecast App, with reasons.

Question 3 [CLO-2] **Marks: 05**
Interpret the initial state, goal test, successor function, and cost function for each of the following problems. Formulate your answer precisely enough to be implemented.
a. Four teams (Red, Blue, Green, Yellow) must each choose a t-shirt color from the set {Red, Blue, Green, Yellow}. A team cannot choose its own name color, and no two teams can have the same color.
b. A 2-foot-tall robot needs to get a package from a shelf that is 6 feet high. The room contains a 3-foot ladder and a 1-foot movable box. The robot can carry one item at a time and can walk, pick up and put down items, stack them, and climb onto them.

Question 4 [CLO-2] **Marks: 05**
Illustrate the graph based on the given edges and explain how Breadth-First Search (BFS), Depth-First Search (DFS), and Uniform Cost Search (UCS) would find a path from Arad to Bucharest.
Graph Edges (with costs): Arad–Zerind(50), Arad–Sibiu(200), Zerind–Oradea(40), Oradea–Sibiu(40), Sibiu–Fagaras(99), Sibiu–Rimnicu Vilcea(60), Fagaras–Bucharest(211), Rimnicu Vilcea–Pitesti(97).

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